

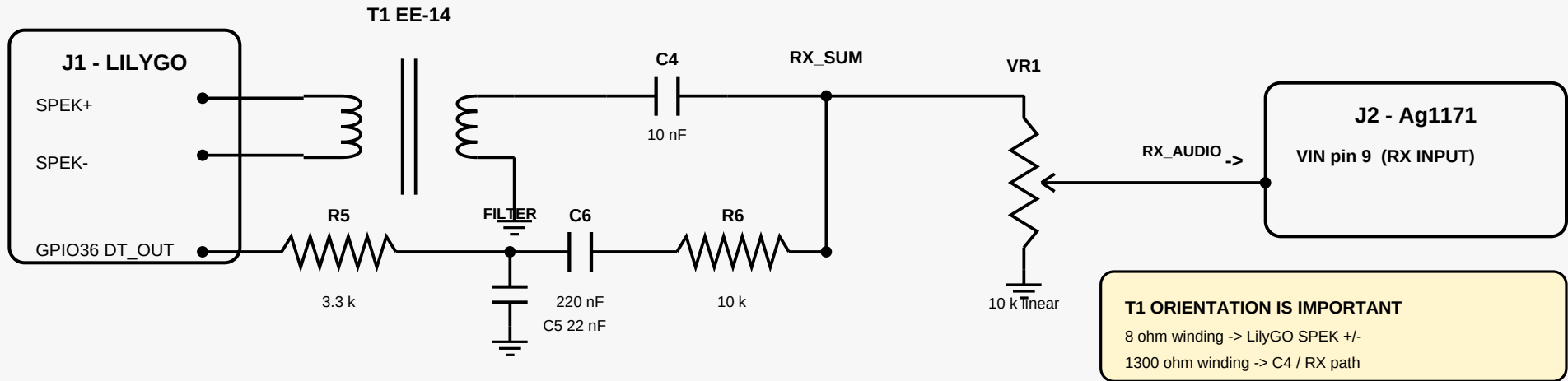
ESP-CellCore - Complete Audio Interface Board

Drawing EC-AUDIO-001

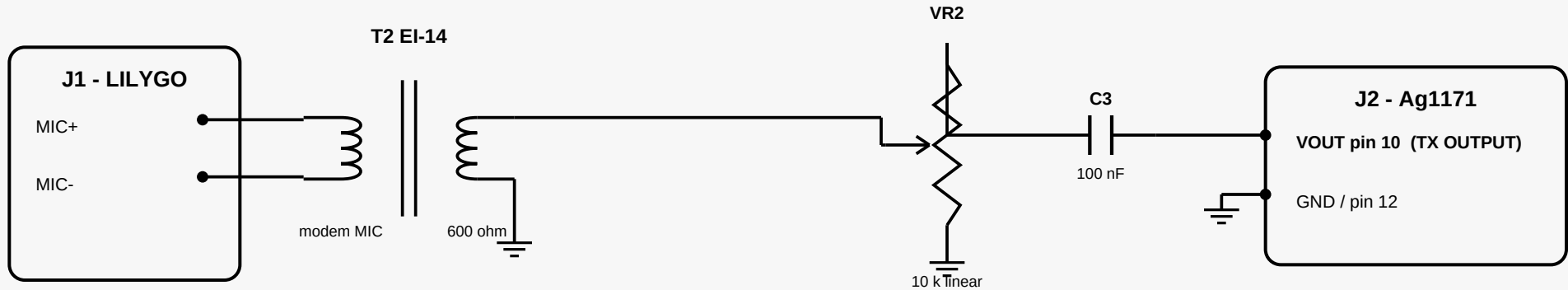
Rev D.1 - corrected Ag1171 audio pin labeling; T1 8 ohm : 1300 ohm; GPIO36 dial-tone injection

2026-07-31

RECEIVE AUDIO + DIAL TONE



TRANSMIT AUDIO



BUILD / TEST NOTES

1. Before first test with the 8:1300 T1, set VR1 near minimum: wiper-to-ground approximately 0 ohm.
2. SPEK+ and SPEK- remain isolated from ground. Only the 1300-ohm T1 secondary is ground referenced.
3. GPIO36 dial-tone path: R5 -> FILTER; C5 to ground; then C6 -> R6 -> RX_SUM. Firmware generates 350 + 440 Hz PWM.
4. T2 remains the existing 600:600 transformer. Adjust VR2 only for rotary-to-cell transmit level / feedback control.
5. Ag1171 PIN 9 VIN feeds cell-to-rotary audio; PIN 10 VOUT feeds rotary-to-cell audio. They are separate paths.